

Web Programming

Rodrigo Martins Pagliares

Last update: 2021/05/31

Web Programming

Rodrigo Martins Pagliares

Web programming foundations

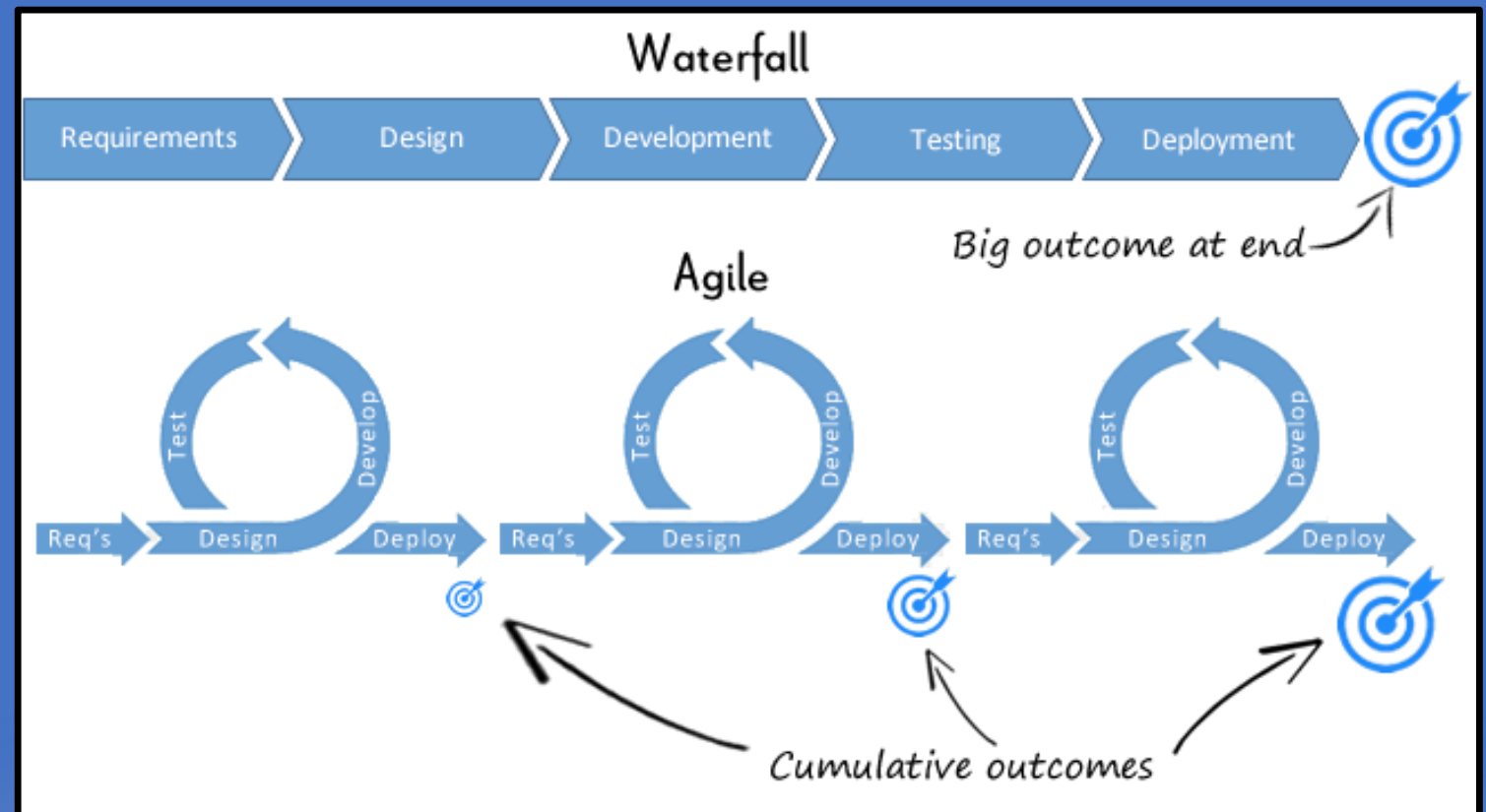
**Development methods,
processes, patterns & practices**

Agile (iterative & evolutive) vs. waterfall development

Lecture: Agile (iterative & evolutive) vs. Waterfall development

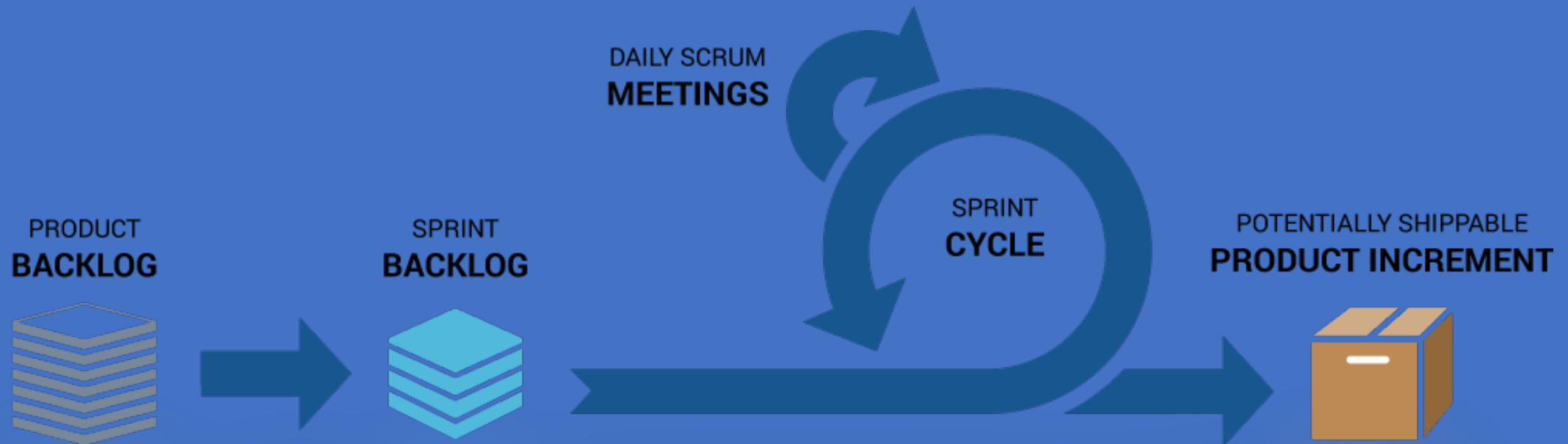
Module: Development methods, processes, patterns & practices

Waterfall vs. agile development



Source: <https://www.facto.com.au/ba/waterfall-vs-agile/>

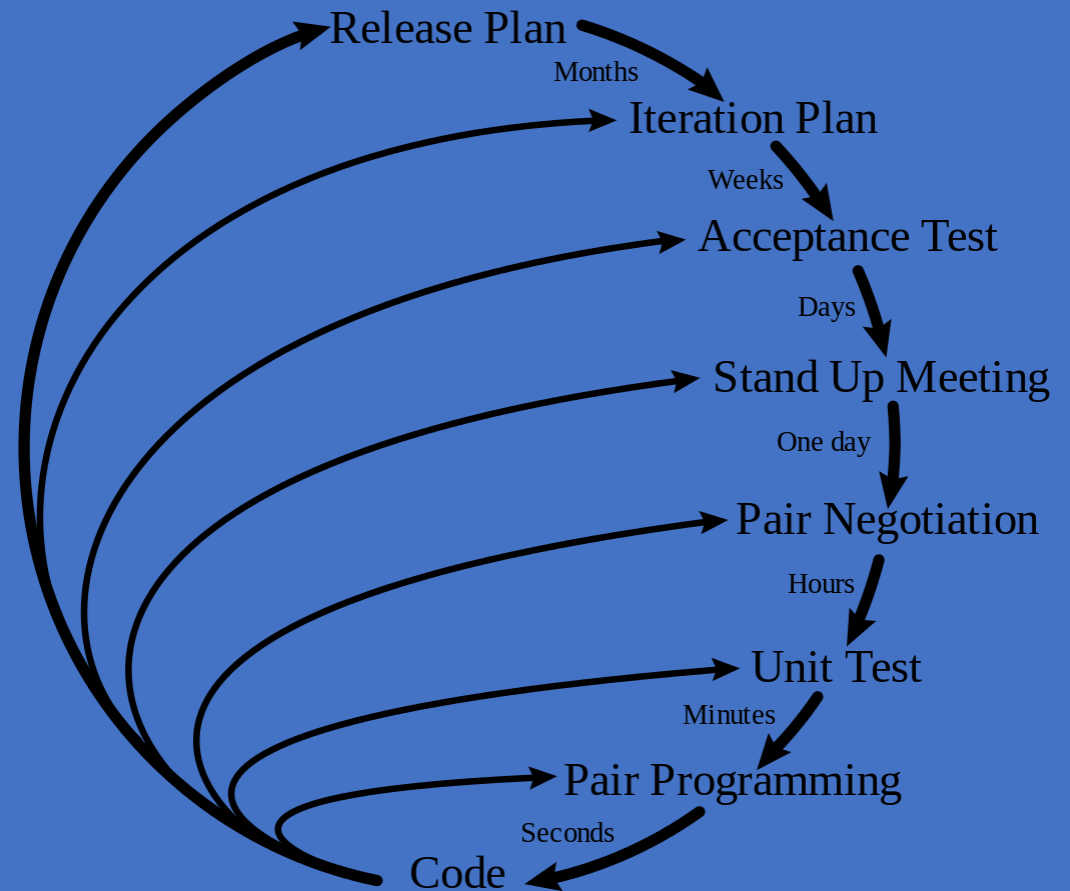
Scrum



Source: <https://www.facto.com.au/ba/waterfall-vs-agile/>

eXtreme Programming

Planning/Feedback Loops



Source: https://en.wikipedia.org/wiki/Extreme_programming

Agile vs. waterfall

- Agile methods like Scrum and XP work by scheduling a series of sprints where you plan, design, build, test, and replan your work every few weeks.
- This is a big improvement from a waterfall lifecycle model, where all the planning is done at the beginning and takes most of the time.

DevOps

Lecture: DevOps

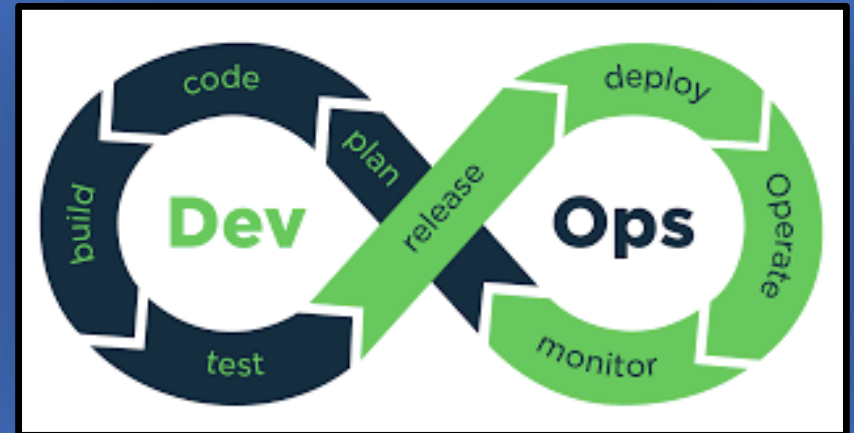
Module: Development methods, processes, patterns and practices

Development + Operations



DevOps

- DevOps combines software development (Dev) with operations (Ops).
- The goal is to shorten the time between writing the code and deployment.



Agile & DevOps

- Agile focuses more on the **communication** between **consumers** and the **product**.
- DevOps focuses on **how** that **software** is **deployed** in the **current environment**.

Development practices

Lecture: Development practices

Module: Development methods, processes, patterns and practices

**The practice
of continuous delivery...**

Continuous delivery

- Today more software is being developed within a processes called Continuous Delivery.
- Many development teams are constantly shipping new features to existing software.
- The software and the success of the code is continuous tested, monitored, and analyzed.



The practice of automated tests...

Automated tests

- **Automated tests** aim at **reducing errors**.
- The use of **automated tests** can be **really useful** because **errors** are **hard to debug** when **applications** start getting **very large**.
- By **automating tests**, you **make sure** that you **constantly check** for **problems** in all **parts** of your **development cycle**.

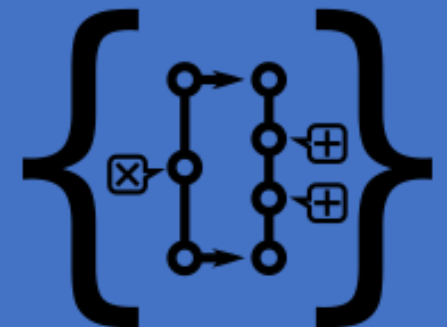


Source: <https://bit.ly/3e1iD2e>

The practice of Version control...

Version control

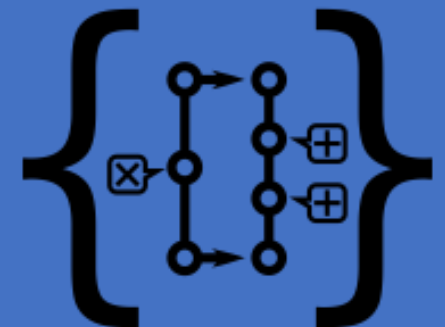
- Suppose **many people** are **working** on the **same piece** of **software** at the **same time**.
- **How do you manage and combine** those **changes** to make a **new version** of the **software**?



Source: <https://bit.ly/37Nj27h>

Version control

- One of the most popular ways to manage software on the web is with **Git**.
- **Git** allows people to work on their own versions of projects and then merge changes together, identifying potential conflicts and providing a way to resolve them.
 - ✓ This is typically referred to as **Version Control**.



The practice of software documentation...

Software documentation

- Good **software** should be **well documented** so that **other developers** and **yourself** can **understand** the **code**.
 - ✓ **Now** and in the **future** !



Source: <https://bit.ly/2HFU7I2>

Teamwork & high-performance teams

Lecture: Teamwork & high-performance teams

Module: Development methods, processes, patterns and practices

Teamwork

- A lot of people have a **misconception** that **developers** are all a **bunch** of **loners**, **working in isolation** with their **headphones**.
- In **reality**, most **developers** work as **part** of a **team**.



Source: <https://bit.ly/3axKtkD>

High-performing teams

- **Developers** make use of **formalized processes, methods and practices** to create **high performing teams**.
- **Agile methods**, for example, focuses on **producing working software** and **regrouping regularly** to make sure that **what** are being **delivered** aligns to **customer needs**.

The pattern Model-View-Controller (MVC)

Lecture: The pattern Model-View-Controller (MVC)

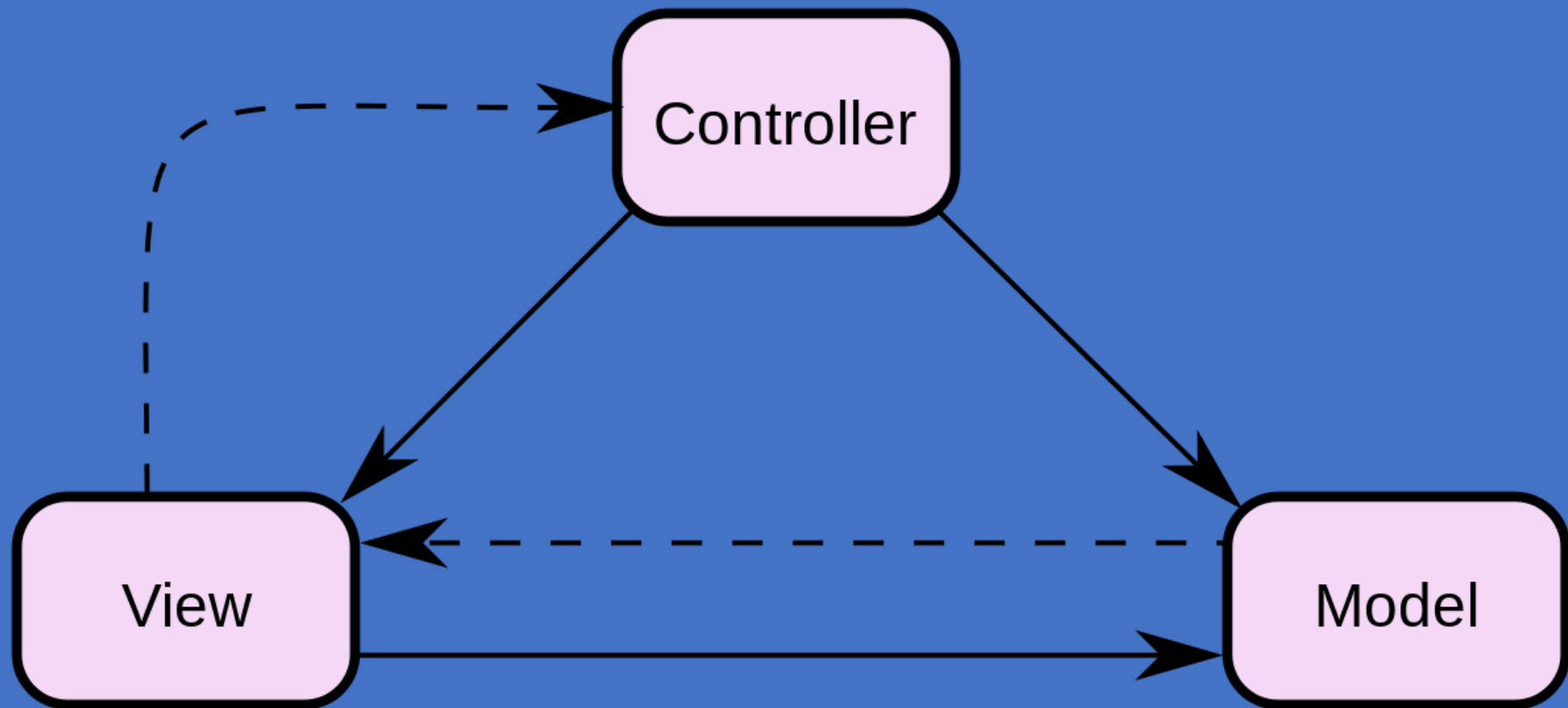
Module: Development methods, processes, patterns and practices

Model-View-Controller (MVC)

- MVC is a **popular design concept** you'll likely come across.
- MVC involves **treating** the **data (model)** as a **separate entity**.
- In MVC, a **template** called the **view** uses the **data**.
- A **code** called **controller** combines the **data** and the **view** together.

The logo for MVC, with the letters M, V, and C in a bold, sans-serif font. The 'M' is green, the 'V' is white, and the 'C' is orange.

Model-View-Controller (MVC)



Source: <https://pt.wikipedia.org/wiki/MVC>

The importance of being a flexible developer

Lecture: The importance of being a flexible developer

Module: Development methods, processes, patterns and practices

The importance of being a flexible developer

- As a **web developer** you will need to learn about these different **methods, processes, patterns and practices**.
 - ✓ but most importantly, you need to be flexible.
- **Developers work with different teams and different approaches, sometimes even within the same company.**
 - ✓ The **developer's ability** to be flexible and embrace different approaches will impact his/her career success and satisfaction.

A **web developer** needs to be **comfortable** with **not just** **how to code**, but also with the **patterns, methods, process** and **practices of how software** is made.

Questions?